

T1. 1p

Inverting op-amp comparator with hysteresis (with positive feedback) without reference voltage: circuit, deduction of the threshold voltages, VTC, waveforms to illustrate the operation.

T2. 1p

Class B power amplifier: circuit; operation for positive and negative input voltage; VTC; waveforms for input and output voltages highlighting crossover distortions; waveforms for output voltage, collector current, collector-emitter voltage and load power. Assume a sinusoidal regime with maximum amplitude of the output voltage.

P1. 2.75p

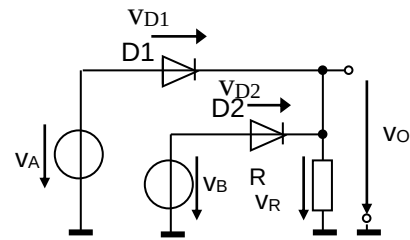
Assume the ideal model for D1, D2.

a)0.5p Deduce the expression $v_O(v_A, v_B)$ and specify the application of the circuit.

b)0.75p Find the values: v_O ; v_{D1} ; v_{D2} if $v_A = -10V$ and $v_B = 5V$.

c)0.5p Find a suitable value for R so that for any v_A, v_B in $[-10V; 10V]$, the maximum forward current through either of the diodes does not exceed 20mA.

d)1p For $v_A(t) = 8\sin\omega t[V]$ and $v_B(t) = -3\sin\omega t[V]$, plot $v_O(t)$ assuming: (1) D1, D2 – ideal; (2) D1, D2 – constant voltage drop model.



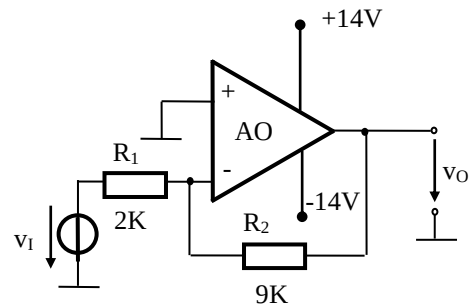
P2. 2.5p Consider AO – ideal.

a)0.75p Find the expression $v_O(v_I)$ – for OpAmp in active region. Find the v_I range for that the OpAmp work in active region.

b)0.5p Plot the VTC $v_O(v_I)$ for $v_I \in [-10V; 10V]$. What is the application of the circuit?

c)0.5p Plot $v_O(t)$ for $v_I(t) = 4\sin\omega t[V]$.

d)0.75p Consider R_2 an open circuit. What is the application of the new circuit? Plot the VTC $v_O(v_I)$ of the new circuit.

**P3. 2.75p**

For OpAmp the maximum output current is $I_{O,AO}=20mA$.

Consider $V_{Ref}=4V$, $v_I=20\pm 2V$, $P=10K\Omega$, $R_1=15K\Omega$, $R_2=4K\Omega$, $R_3=6K\Omega$.

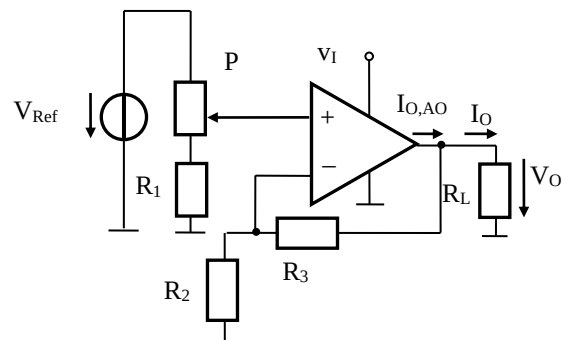
a) 0.5p What is the sign of circuit feedback and what is the application of the circuit?

b) 0.75p Find the range of the output voltage V_O .

c) 0.5p Find the minimum value of load resistance R_L so that the application of the circuit remains the one determine to point a) .

d) 0.5p For the tap of the potentiometer in the middle position compute the maximum power dissipation to the output if the load resistance is $R_L=1.5K\Omega$.

e) 0.5p Consider R_2 an open circuit. Is the range of V_O modified? If yes, what is the new range of V_O ?



T1. 1p

Noninverting op-amp amplifier: circuit, deduction of the differential voltage, deduction of the gain, input and output resistances, VTC, range of the input voltage for the active region, waveforms to illustrate the operation including the saturation of the amplifier.

T2. 1p

Op-amp voltage regulators. Solutions for: 1) $V_O > V_{REF}$; 2) $V_O < V_{REF}$; 3) adjustable V_O , $V_{Omax} > V_{REF}$. For each solution: the circuit and expression of V_O .

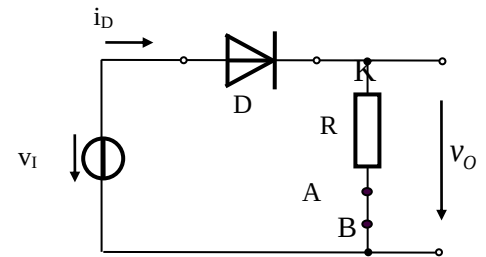
P1. 2.75p Consider D – ideal.

a)0.75p Deduce and plot the VTC $v_O(v_I)$. On the VTC specify the state of the diode.

b)0.5p Plot $v_O(t)$ for $v_I(t) = 8\sin\omega t$ [V].

c)0.75p Connect a bias source between A and B points, $V_{BIAS} = -3V$. Deduce and plot the VTC $v_O(v_I)$ for the new circuit, assuming: (1) D – ideal; (2) D - constant voltage drop model.

d)0.75p For the circuit at point c), consider D – ideal and $v_I(t) = 8\sin\omega t$ [V]. Size the R resistor so that the maximum current through D will be $I_{D, max} = 200mA$.

**P2. 3p**

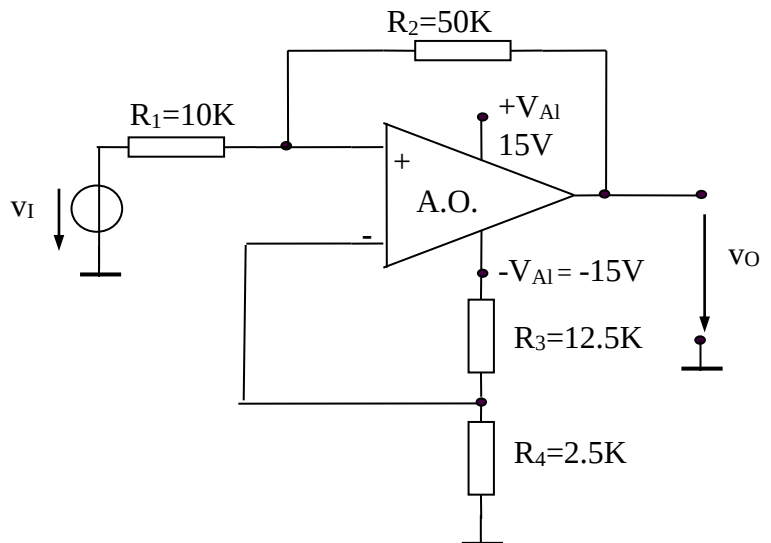
OpAmp – ideal.

a)0.5p What is the application of the circuit? Plot, qualitatively, the VTC $v_O(v_I)$ of the circuit.

b)1p Find the expressions and values of the following parameters of the VTC: V_{ThL} , V_{ThH} , V_{OL} și V_{OH} ? Redraw the VTC $v_O(v_I)$ according to the numerical values obtained.

c)0.75p Plot $v_O(t)$ for $v_I(t) = 10\sin\omega t$ [V].

d)0.75p Consider R_2 an open circuit. What is the application of the new circuit? Plot the VTC $v_O(v_I)$ of the new circuit.

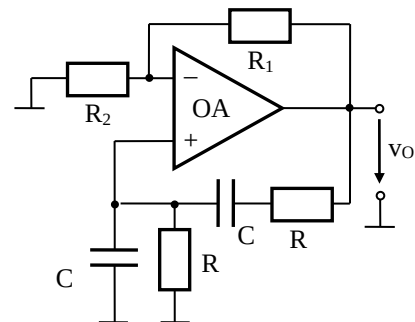
**P3. 2.25p**

For this circuit $R_2 = 6.8K\Omega$ and $C = 10nF$.

a) 0.5p Assume that the Barkhausen condition is fulfilled. Find the values for the resistances R to obtain oscillation on $f_0 = 8$ KHz.

b) 0.5p Deduce the value of R_1 to fulfill the oscillation (Barkhausen) condition.

c) 1.25p Using diodes modify the circuit to assure an automatic gain control. Assuming $r_d = 1.5K\Omega$ of the conducting diode in the moment when the Barkhausen condition is satisfied, size all the resistors in circuit. Verify if the start-up condition is also satisfied.



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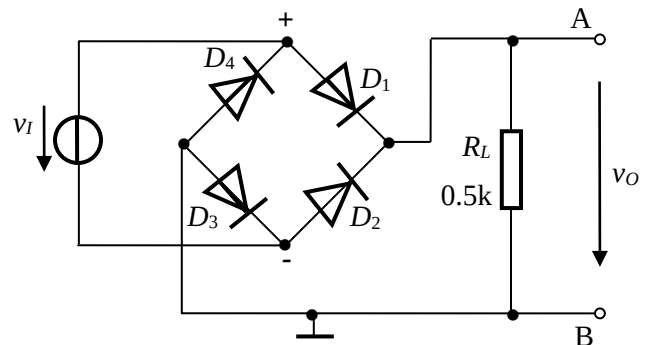
P1. 2.25p

Consider $v_I(t)$ a sine wave, 20V amplitude and 500Hz frequency and for diode – the voltage drop model with $v_D=0.6V$.

1p a) What is the application of the circuit? How do $v_O(t)$, the current through D_1 - $i_{D1}(t)$ and the current through D_2 - $i_{D2}(t)$ look like?

0.75p c) Consider that a capacitor C is connected at the output (between A and B points). Draw the new circuit. What should be the value of the capacitor so that the output ripple $\Delta v_O < 2.5V$?

0.5p d) How does $v_O(t)$ and the currents through D_1 and D_2 look like if the diode D_4 is considered an open circuit?



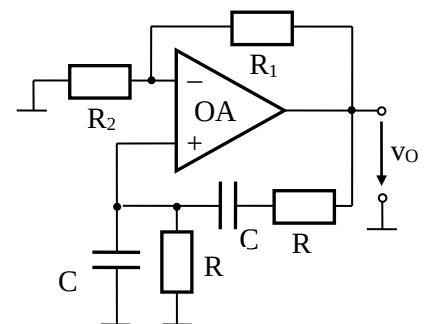
P2. 3p

For this circuit $R_2=5.6K\Omega$ and $R=8K\Omega$.

0.75p a) Assume that the Barkhausen condition is fulfilled. What are the values of capacitor C to obtain oscillation on $f_0=5KHz$?

0.75p b) Deduce the value of R_1 to fulfil the oscillation (Barkhausen) condition.

1.5p c) Using diodes, modify the circuit to assure an automatic gain control. Assuming $r_d=2.5K\Omega$ of the conducting diode in the moment when the Barkhausen condition is satisfied, size all the resistors in circuit. Verify if the start-up condition is also satisfied.



P3. 2.75p

Consider $R=1K\Omega$, $R_L=20\Omega$.

0.5p a) To what class of output stages does the circuit belong? Justify your answer by evaluating the conduction periods of T_1 and T_2 for a sinusoidal input voltage $v_i(t)$, with 5V amplitude.

0.75p b) Deduce and plot VTC $v_O(v_i)$. Specify the (on) or (off) states of the transistors on the plot.

0.75p c) Plot $v_O(t)$; $i_{C1}(t)$; $v_{CE1}(t)$, for $v_i(t)=12\sin\omega t$ [V]?

0.75p d) For $v_i(t)=12\sin\omega t$ [V] compute: the average a.c. power transmitted to R_L , the average power consumption from the two power supplies, the efficiency.

